



# Incident Response Simulation Report – SystemTron

## Cybersecurity Internship

-  **Name:** Somashekhar Hiremath
-  **Project Title:** Simulated Cybersecurity Incident Response
-  **Tools Used:** Kali Linux, Metasploit, Wireshark, ClamAV, chkrootkit
-  **Target Environment:** Metasploitable 2 VM

### 1. Objective

To simulate a cybersecurity incident and develop real-world incident detection, response, and recovery skills using commonly available tools and techniques.

### 2. Simulated Incident Setup

- A reverse shell attack was simulated from **Kali Linux** to **Metasploitable 2** using Metasploit.
- The vulnerable **vsftpd 2.3.4** service was exploited to gain shell access to the target machine.

#### Commands Used:

```
msfconsole
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOST <Target_IP>
run
```

### 3. Incident Detection

Monitored the target system for signs of compromise:

| Command        | Purpose                                 |  |
|----------------|-----------------------------------------|--|
| who            | Check currently logged-in users         |  |
| last           | View login history                      |  |
| ps aux         | List running processes                  |  |
| netstat -antup | Check open ports and active connections |  |

| Command                   | Purpose         |                                   |
|---------------------------|-----------------|-----------------------------------|
| `cat<br>/var/log/auth.log | grep "Failed"`` | Detect brute-force login attempts |

Findings:

- A suspicious shell session was active.
- Logs indicated unauthorized login attempts.
- A reverse TCP connection was observed through netstat and Wireshark.

 4. Containment & Eradication

Actions Taken:

- Disconnected the Metasploitable VM from the network.
- Identified and killed the malicious shell process.
- Removed suspicious files from the /tmp directory.
- Scanned the system for malware using ClamAV and chkrootkit.
- Verified there were no unauthorized user accounts.

Commands Used:

```
ps aux
kill <PID>
ls /tmp
rm -f /tmp/<malicious_file>
clamscan -r /home
chkrootkit
cat /etc/passwd
sudo userdel <rogue_user>
```

 5. Root Cause Analysis (RCA)

- **Cause:** The system was running **vsftpd 2.3.4**, which is known to have a backdoor vulnerability.
- **Weaknesses:**
  - Outdated software
  - Open FTP port exposed without firewall restrictions
  - No monitoring or alerting system

## 6. Recommendations

- **Disable vulnerable services** like FTP or replace with secure alternatives (e.g., SFTP).
- **Update and patch** all services regularly to avoid exploitation.
- **Use firewalls** to restrict access to sensitive ports.
- **Deploy IDS/IPS** to monitor real-time activity and trigger alerts.
- **Harden systems** by removing unused accounts and disabling default credentials.

## 7. Lessons Learned

This hands-on experience helped me:

- Understand the importance of early detection.
- Practice real-world containment and eradication techniques.
- Learn how to document findings and perform a root cause analysis.

## Supporting Evidence:

- Screenshots of Metasploit exploitation
- Shell access detection using `netstat` , `ps` , and logs
- Removal of threats and user cleanup
- Wireshark capture during incident